Chapter 4

Pharmacy Management

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Learning Objectives

Objective 4.1 Outline the major principles in managing a pharmacy.
Objective 4.2 Overview the purchase, inventory control, storage, drug distribution, compounding, and disposal of drugs and related products.
Objective 4.3 Illustrate a model pharmacy and detail the facilities required.
Objective 4.4 Overview pharmacy practice setting and responsibilities of professionals.
Objective 4.5 Explain advantages and limitations of pharmacy automation.

OBJECTIVE 4.1. OUTLINE THE MAJOR PRINCIPLES IN MANAGING A PHARMACY

The effective management of a pharmacy requires much more than what is normally seen as the task a pharmacist performs, namely dispensing drugs and consulting patients or providers. It requires skills to effectively manage personnel, inventory, and various licensing formalities including documentation, finances, and facilities. Pharmacists are required to know how to manage stock levels, make provisions for alternatives, know how to budget and manage the finances of a pharmacy, and, above all, ensure patient safety and drug optimization.[1]

It is well known that healthcare costs are a significant and rising expense for countries throughout the world. In the United States, costs of care are particularly worrisome, as some predict healthcare spending will account for 19.7% of the economy by 2026.[2] The same authors estimate that medications will account for nearly 11% of healthcare spending by 2026,[2] reinforcing the need for the cost-effective use of medications. For many health systems, pharmacy departments are being asked to evaluate and address rising healthcare costs through the judicious use of medications.[3] To be effective in this environment, pharmacy leaders must have a firm understanding of basic management principles and must develop and implement pharmacy services with highly trained personnel to help improve affordability and quality.

The major principles in managing a pharmacy are financial management, personnel management, and regulatory compliance.

Financial Management

Besides being a patient care location, pharmacies are also businesses and need to manage financial resources effectively. Financial management for a given period at its most basic consists of balancing revenue against costs by developing and implementing a budget or financial plan. Sustainable pharmacies must take in at least as much money as they spend. In some countries, hospitals are state-run, and revenue-generating business practices may not apply. Instead, hospital pharmacies often resemble cost centers and as such need to apply business principles to utilize resources such as drugs, staff, and utilities efficiently and within budget restrictions.

Costs for pharmacy managers come from many sources (Table 4.1). Some of these costs (such as utilities) are budgeted separately in hospitals and retail pharmacy stores, and the pharmacy manager may not have specific budgets to consider for lighting and temperature. The pharmacy manager, however, is responsible for drug costs, personnel, and revenue cycle management. Drug costs represent the highest budgeted expense for a pharmacy of all potential categories. Managing drug inventory has a significant impact on how effectively one manages drug costs and is discussed below.

Clinical Pharmacy Education, Practice and Research. https://doi.org/10.1016/B978-0-12-814276-9.00004-0
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TABLE 4.1 Common Costs for Pharmacies

Medication costs	• Cost to purchase medications from manufacturers or wholesalers
Personnel costs	• Salary • Benefits (health insurance, life insurance, dental/vision insurance, vacation time, disability, protected medical or family leave, etc.)
Automation/technology	• Purchase and maintenance fees for computers, automated dispensing cabinets, dispensing carousels, intravenous admixture robotics, medication repackaging machines, pill counters
Utilities/infrastructure	• Lighting • Heat/air conditioning • Water/sewer • Rent • Telephone/Internet
Regulatory	• Licensing and accreditation fees • Liability insurance for employees
Supplies	• Medication labels • Pill bottles/caps • Bags • Pens/paper • Desks/chairs • Syringes/needles • Scales • Compounding supplies

Other sources of expense for a pharmacy include various automation and technology expenses. Today, the use of electronic health records, computerized dispensing software, and vast amounts of medical data and references has rendered Internet and computer resources a necessity for effective and efficient practice. In some hospitals, the use of technology is so pervasive that pharmacists are provided individual laptops or mobile workstations to be more effective while rounding through patient care areas. Additionally, pharmacists in larger hospitals are often provided some type of paging device or telephone at which they could be easily reached for urgent communications. Each of these resources has an initial purchasing price and in many cases a monthly fee, either for maintenance or continued access and use of the system.

On a larger scale, dispensing automation represents a significant expense for some pharmacies. With purchase prices ranging from thousands to hundreds of thousands of dollars, automation ranging from pill counters to sterile compounding robots to automated dispensing cabinets to medication storage carousels requires significant planning and preparation to yield benefit. In most cases, the technology is justified financially by improving dispensing accuracy, safety, decreasing the footprint required for medication storage by taking up less total space, meeting regulatory requirements for secure storage of medications, or improving the efficiency of the dispensing process (thereby saving salary cost). A pharmacy manager must carefully assess the medication use process to determine where a piece of automation may provide benefit and assess and manage costs associated with purchase and maintenance of such a device.

Computers and automation or technology are generally simple expenses to plan for—purchase prices are known in advance, and costs do not change based on how much or how little each component is used. Such expenses are considered "fixed." Expenses such as medications and any supplies needed for dispensing (pill bottles, labels, printer ink, bags, etc.) are considered "variable", as the total amount spent by the pharmacy will vary with the number of prescriptions dispensed or medication orders processed. Both types of expenses must be considered by a pharmacy manager when planning annual budgets.

Revenue

Although all pharmacies must address costs, not all pharmacies generate revenue directly. In many outpatient models, pharmacies may charge patients a fee based on the type and quantity of medication being dispensed. This practice varies substantially across the world, with some countries providing medication coverage, whereas others, such as the United States, utilize public and private insurance companies to cover a portion of the expense, with the remainder paid for by the patient. Depending on the practice setting, pharmacy managers may have some ability to set or negotiate pricing with

various payers. In private-pay situations, where the patient is responsible for the whole expense, pharmacies have significant flexibility in setting prices. In settings where insurance companies or government payers handle payment, there is generally more limited opportunity for a pharmacy to negotiate prices and less flexibility in adjusting those prices outside of the contract period. For example, in the United States, it is common for a medical insurance company to negotiate reimbursement for pharmacist services. These services include some accounting for the cost of the medication plus a set fee for the time of the pharmacist and support personnel (often called a "dispensing fee"). The collection of these fees, whether negotiated with insurance companies or set independently by the pharmacist, represents the source of revenue for the pharmacy and thus a process that needs scrutiny by the pharmacy manager.

In hospital pharmacies, there is less opportunity to generate revenue. In many countries, including the United States, inpatient pharmacy services are considered a component of the hospital stay and are thus billed and reimbursed as a component of overall care. Using more expensive drugs without achieving better outcomes decreases available funds for the health system. Pharmacy departments are responsible for assuring cost-effective use of medications to maximize the amount of revenue available for use in improving or expanding hospital services. This goal is achieved through the judicious application of pharmacist training and experience to guide prescribing practices toward the most cost-effective therapy.[4] The role of pharmacist position design, recruitment, and management is an important one for the pharmacy manager and is discussed more fully below.

Budgeting

As mentioned previously, a significant role for the pharmacy manager is planning for expenses and revenues to ensure the appropriate financial performance of the pharmacy. Budgeting is the process of planning expenses and revenues for an upcoming period. As many variables impacting pharmacy expenses and revenues change over time, budgeting is often completed annually through a "budget cycle" (Fig. 4.1). The basic components of a budget cycle include planning, review and approval, implementation, and evaluation.

Planning

Budgeting for healthcare services is commonly achieved through an incremental budgeting process, wherein the budget from the prior budget cycle is reviewed and revised to reflect anticipated trends, new services, or other changes. To begin the budgeting cycle, a pharmacy manager reviews the evaluation from the prior budget cycle and considers emerging trends (new medication approvals, new studies influencing use of existing medications, new reimbursement models affecting the pharmacy, new generic medication availability, ending technology service contracts) and service gaps (opportunities to add services [and thereby workload and potentially personnel expense]). Additionally, pharmacy managers must consider whether the volume of existing work might change (either through the emergence of new technology or through increased patient volumes such as the addition of a new wing of a hospital or a new insurer contract with a retail pharmacy system). Once the pharmacy manager has considered emerging trends, he or she must estimate the impact on workload, expense, and revenue and account for these changes in the budget.



FIGURE 4.1 Basic budget cycle.

One challenging opportunity with high potential risk and reward is the addition of pharmacy personnel to begin new service. As described above, the changing landscape of healthcare away from a fee-for-service design to an outcomes-driven reimbursement model opens new doors for pharmacist involvement and ownership of the medication use process, including prescribing of medications. During the budget cycle, a pharmacy manager must assess which new services could be used to achieve improved outcomes and reimbursement for the organization, preparing a business case that outlines the role of such services, the quality impact, and resulting financial impact. These cases are designed to describe the return on investment (ROI) of a pharmacist (for every dollar spent on the pharmacist, how many dollars does the system save or gain over a defined period) (Table 4.2). Once the budget is described and any new services have an accompanying business case, the budget is submitted for review and approval.

Review and Approval

The review and approval process of a pharmacy budget will vary significantly by practice setting. In a large health system, for example, budgets from all disciplines (pharmacy, nursing, physicians, respiratory therapists, etc.) are reviewed alongside one another. The first priority for approval is given to maintaining current services and addressing increased visit volumes by patients. Once this review is completed, the team reviews new business case opportunities (such as those submitted for new pharmacy services). There are two common calculations used to help compare cases: ROI and breakeven. While ROI focuses on the amount of financial gains or savings achieved through the investment, breakeven focuses on how quickly initial start-up costs are recovered through those gains or savings. Business cases with high ROI, fast breakeven, and low overall risk have the strongest chance of approval. Once approved, final budgets are distributed back to the manager for implementation.

TABLE 4.2 Example of a Pharmacy Business Case Template

Example Business Case Template
1. Executive summary **a.** Proposal **b.** Background **c.** Benefits **d.** Financial impact **e.** Conclusion **f.** Return on investment (ROI) table (brief) **2.** Background **a.** Description of problem **b.** Evidence of current state **c.** Solutions implemented elsewhere **3.** Proposal **a.** Full-time equivalent/resource needs **b.** Benefits **4.** Potential impact **a.** Financial impact **b.** Nonquantifiable benefits **5.** Alternatives (with financial impact and pros/cons list)—*often removed in final case* **a.** Brief review of other solutions considered and key reasons they are not preferred **6.** Implementation plan **a.** Who does what and when **7.** Assumptions **a.** Describe trends or values assumed static in benefit/risk calculations **8.** Sensitivity analysis—often removed in the final case or embedded in the description **a.** Pick assumptions that have the greatest impact on ROI and some risk of variability. Assume worst-case scenario and evaluate benefit/risk in these cases. Alternatively, if worst case is unknown, select values representing minimum necessary for benefits to equal risks and determine the likelihood of worse performance. **9.** Risks and mitigation plans **a.** If the significant risk is identified during sensitivity analysis, describe monitoring plan to detect early and steps to take to mitigate risk if detected. **10.** Exit Strategy **a.** If risk comes true, how to minimize adverse impact on the organization

Implementation

The manager is responsible for implementing the budget as outlined and assuring appropriate resource management throughout the budget year to stay on target. Budgets represent financial plans for an organization, providing a guidance on decisions impacting financial performance. During implementation, managers can begin to hire employees for newly approved positions, can schedule and complete needed renovations, or purchase and implement new automation or technology.

Evaluation

Budgets represent a plan of action for the pharmacy department, and both overestimation and underestimation of expenses and revenues create lost opportunity. Although one can easily see how spending more than planned is negative (the pharmacy or hospital may or may not have sufficient money on hand to handle significant underestimations of expenses in a budget), overestimating expenses can lead to lost opportunity by tying up funding in planned expenses that never gets spent. In such cases, opportunities to improve facilities or automation and technology or add a new service may be set aside, creating a delay in quality or efficiency advancements. To safeguard against this potential, managers frequently assess variance from the budget and identify causes. For example, a manager may review monthly productivity reports and identify staff is completing far more work than planned, leading to potential overtime expenses and burnout. Identifying the cause (perhaps increased admissions above what had been anticipated) can help identify solutions to head off negative consequences of the budget deviation. Managers often document these deviations in "variance reports," which are used to help explain why or why not a budget was accurate, offering the opportunity to improve accuracy during the next budget cycle.

Personnel Management

Another common source of expense for a pharmacy is staff salaries. Pharmacies may employ a wide array of positions, including pharmacists, pharmacy technicians, clerical staff, administrative support staff, data analysts, and informatics personnel. Salaries may be paid out at a set amount regardless of hour worked (called "salaried" employees) or at a set amount per hour of work (called "hourly" employees).

Management strategies for salaried and hourly employees generally revolve around the principle of a full-time equivalent, or FTE, representing a standard 40-h work week in the United States. Pharmacy managers must decide whether a particular position is most appropriately hourly or salaried. Many pharmacist positions are considered salaried especially when responsibility is assigned for a particular group of patients. For example, an infectious disease pharmacist may participate in patient care rounds with the infectious disease consult service. This pharmacist may be staffed as a salaried employee because the volume of patients seen may require more than a typical 8-hour day in some cases and may require a bit less in others. This variable work schedule lends itself well to a salaried role because the pharmacy manager still may accurately predict (and therefore budget) the salary. In an hourly position, such accuracy is obtained through use of scheduled shifts. For example, an employee may start work at 07.00 and finish at 15.30 daily. Paid at an hourly rate, this schedule yields a consistent expense the pharmacy manager can predict and plan for. If the workload was highly variable, additional shifts may be needed (for example, 07.00−15.30 and 14.30−23.00). To determine which schedules are needed to address the workload, pharmacy managers rely on productivity monitoring and benchmarking.

Productivity

As pharmacists work to improve the care of their patients, pharmacy leaders work to ensure the availability of pharmacists to meet patients' needs. To do so, pharmacy leaders rely on information about the productivity of the pharmacists. Productivity is a ratio of outputs (work done) to inputs (pharmacist time). Outputs in pharmacy practice can be expressed in many ways, ranging from the number of orders verified by the number of patients seen in an ambulatory clinic in the volume of pharmacokinetic monitoring completed. Other measures of outputs serve as surrogates for total pharmacy practice. For example, many productivity calculations include the number of patient admissions or ambulatory visits or the severity of illness of patients the pharmacist cares for.[5] As these surrogate measures increase or decrease, pharmacy managers anticipate an increase or decrease in total work done to provide care. Inputs for productivity assessments are often expressed as some function of time. Time could be expressed as total pharmacist hours worked or as pharmacist FTE (the number of pharmacists that could be hired if all worked 40 h per week). In one way or another, inputs in pharmacist productivity calculations represent an expense for the healthcare organization, with the outputs representing the value or benefit resulting from that expense. For healthcare organizations to be successful, they must find a sustainable balance between outputs and inputs.

Pharmacy leaders and governments pay close attention to pharmacist productivity, as it directly impacts their goal to ensure availability of pharmacists for patient care needs. If productivity decreases, the healthcare organization may conclude that there is insufficient work to justify hiring more pharmacists and may even choose to reduce the total number of pharmacists it employs. If productivity increases, it may represent the increasing workload for pharmacists, creating additional pressure to complete all the necessary work in a shorter amount of time—eventually, this trade-off could lead to less effective pharmacy practice. To help identify reasonable expectations for productivity, pharmacy leaders benchmark productivity calculations against other pharmacy teams in the health system (internal benchmarking) or other pharmacy teams outside the health system (external benchmarking).[5] The success of benchmarking relies on comparing similar processes—if two pharmacy teams have very different approaches to patient care, benchmarking one against the other will not yield useful information. Although there is no gold standard for reporting productivity, some common methods are used to help ensure comparisons are made similarly across institutions.

Because much of what advanced pharmacy practitioners do is a cognitive service instead of a discretely identifiable action, measuring pharmacist productivity is challenging. Common productivity reporting and benchmarking systems may not adequately reflect specific clinical functions of interest to health system leaders. Early reports evaluated clinical pharmacist activities, including patient care rounding, chart reviews, responding to drug information requests, and performing literature searches. Some assessments also included distributive functions such as obtaining first doses from a central pharmacist, ordering medications, and checking unit-dose carts. Given the capabilities of health information systems at the time, simplifying calculations made quantifying clinical pharmacist activity feasible. As the speed and efficiency of electronic data collection and analysis have increased, pharmacists can apply more complex productivity assessment techniques.

Work productivity of pharmacists is studied in medical and surgical patients, high- and low-risk patients, orders for specific medication classes, emergency resuscitation, and medication cost management.[6–8] In comparing time needs for medical and surgical patients and high- and low-complexity patients (identified through diagnosis-related groups), a significant difference was noted between high- and low-complexity patients for all documented activities except drug information request. Significantly more time was required for medical patient care than surgical patient care.[6] In separate studies, pharmacy teams utilized input from pharmacists to weigh various pharmacist actions (order verification, code response, etc.) based on the perceived complexity of cognitive assessments needed to complete the action.[7,8] More complex activities were assigned higher values to adjust for the time needed to assess available information appropriately.

Because productivity monitoring has an indirect effect on patient care, all pharmacists should have a basic understanding of how productivity is measured and how it is used to make decisions about assigning available pharmacist services. To ensure success, pharmacists should review productivity measures used by their employers to determine appropriate staffing levels and provide feedback to identify the most accurate assessment possible in a given health system. Only by understanding and influencing productivity assessment can one be sure that sufficient time is protected to achieve the outcomes discussed previously.

Job Design

In developing a new position or service, a pharmacy manager must consider more than salary to manage the team effectively. A pharmacy manager also has responsibility for job design, recruitment, training, schedules, and performance management.

One component of pharmacy management is consideration of the job design of the various pharmacy employees to ensure that the overall structure of the department of pharmacy meets the needs of patients, providers, and other customers. Although there are many ways to organize thoughts around a new position, one method includes working backward through the Donabedian model (Fig. 4.2).[9] Beginning with outcomes of interest, the pharmacy manager identifies as clearly as possible what the role should achieve, then outlines how the pharmacist would go about achieving those objectives and considers what core structural elements the position requires (such as training, licensure, advanced certifications, and years of experience). These concepts come together as a position description, which is used to set expectations of job performance and as a communication tool during recruitment.

Recruitment

When a position becomes available, the pharmacy manager must oversee recruitment and hiring of a new employee. Applications are received and candidates are screened to ensure they meet minimum requirements outlined in the position

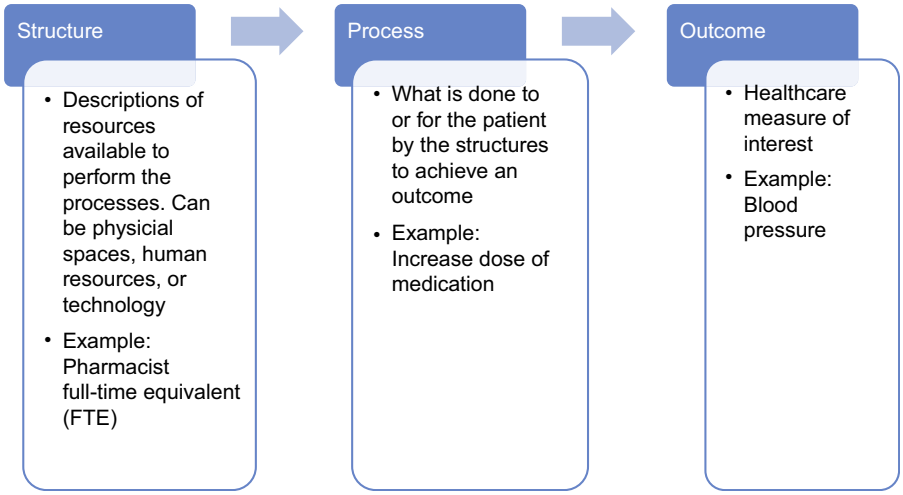


FIGURE 4.2 Donabedian model.

description. Appropriate candidates are contacted, and an interview is scheduled. Managers utilize some interview techniques to determine which candidate seems to be a great fit for the needs of the role. Managers use combinations of behavioral and situational questions to determine how the candidate will likely respond to various scenarios they are likely to encounter in the position. Behavior-based interviewing focuses on how the candidate responded to similar scenarios in the past ("tell me about a time when …."), anticipating future behavior to mirror previous behavior. Situational questions are used when the candidate may not have had specific experiences as they would encounter in the role. These questions are future-focused ("how would you respond if …"). Interviewing is such a complex skill to develop that several popular publications exist around the topic.[10]

Training

Once a job offer has been made and accepted, the manager must oversee the initial training period for the new employee. This period covers a large number of policies and procedures, instructing the new pharmacist on how the current team ensures compliance with applicable laws and regulations, as well as how care is delivered and documented. The pharmacist is taught how to use the specific electronic health record and other automation or technology the pharmacy may utilize, and gradually the employee becomes familiar with the various roles and functions within the team.

Performance Management

During orientation, pharmacists should also have clear expectations of performance set. In many cases, performance management for pharmacist presents significant challenges. Although performance management practices is another large and complex topic, many managers have found the concept of a "balanced scorecard"[11] to be an effective strategy for developing a well-rounded set of performance expectations for pharmacists. As described by Kaplan and Norton, the balanced scorecard outlines performance measures from the financial, business, innovation/learning, and customer perspectives. In healthcare, this model can be adapted to include measures of drug cost and revenue, productivity, education and research, and patient or healthcare worker satisfaction (Fig. 4.3). Developing objective and effective performance measures is a challenge in pharmacy, as much of the service is completed as a cognitive function (challenging to measure) and outcomes are not always available to measure (such as prevention of adverse effects by adjusting a dose or decreasing mortality by adding or revising a medication regimen). Nonetheless, a major role of a pharmacy manager is evaluating performance and providing feedback to employees to help improve effectiveness and efficiency. Pharmacy managers utilize available objective data combined with subjective input from coworkers, technicians, and other medical professionals. This type of all-around review is known as a 360-degree evaluation.[12]

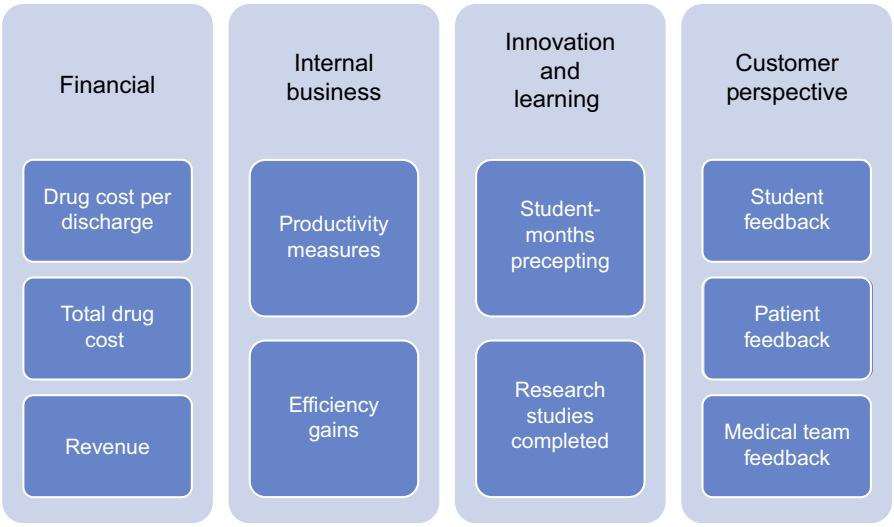


FIGURE 4.3 Balanced scorecard for pharmacist performance. *Adapted from Kaplan RS, Norton DP. The balanced scorecard—measures that drive performance.* Harv Bus Rev. *1992;70(1):71—79. http://www.ncbi.nlm.nih.gov/pubmed/10119714.*

Regulatory Compliance

Another important role of the pharmacy manager is to ensure compliance with applicable laws and regulations. In the United States, a myriad of regulatory bodies oversees the structure and function of a pharmacy team. At the federal level, the Food and Drug Administration in the United States oversees which pharmaceutical agents can be bought and sold for patient use and what labeling must be included. Accrediting bodies such as the Centers for Medicare and Medicaid Services utilize their status as a major payer to enforce process and quality requirements. The Drug Enforcement Agency also has specific requirements around prescribing, purchasing, storing, and dispensing controlled substances with abuse potential. Each state has its own Board of Pharmacy, which can enact rules governing how pharmacists practice. There are many more examples of local, state, and federal agencies with authority to govern pharmacy practice, leading to small but important distinctions across the United States. These groups will provide periodic inspection of pharmacy practice areas to ensure compliance with the applicable rules. If a pharmacy is found to not comply with various requirements, punishments can range from fines to revocation of license, essentially shutting down the pharmacy and potentially removing the ability for the pharmacy manager to practice or operate a pharmacy again. As such, pharmacy managers must remain knowledgeable on statutes and rules affecting pharmacy practice and must enact policies and procedures for staff to follow, ensuring compliance with the law and continuation of the practice.

OBJECTIVE 4.2. OVERVIEW THE PURCHASE, INVENTORY CONTROL, STORAGE, DRUG DISTRIBUTION, COMPOUNDING, AND DISPOSAL OF DRUGS AND RELATED PRODUCTS

The pharmaceutical industry manufactures drugs and other healthcare products. They are distributed to the retail community or hospital pharmacies through pharmaceutical distribution channels. The integrity of these distribution systems is important to avoid infiltration of counterfeit products in the supply chain, as well as to maintain the correct transport and storage conditions.[13] The pharmacy manager is also responsible for oversight of the pharmacy's purchasing practices. Here, we will review basic principles of inventory management.

Methods in inventory control such as ABC analysis and VED analysis are important measures in maintaining optimum stock levels. ABC analysis, otherwise known as "Always Better Control," is a method of controlling stock according to Pareto's principle of "vital few and trivial many." This principle states that approximately 10% of hospital medications are responsible for 70% of the budget (A). The next 20% (B) consume 20% of the budget, whereas the remaining 70% of the products (C) account for the last 10% of the budget. It is therefore extremely important to manage the 10% of medications responsible for 70% of the budget (A) (Table 4.3).[14]

TABLE 4.3 ABC Analysis

Item	Items (%)	Money Value (%)
A	10	70
B	20	20
C	70	10

VED analysis is based on the criticality of the product, where "V" determines the items that are vital to the hospital's system without which they cannot function, and "E" are the essential items without which the hospital's quality of service might be affected. "D" stands for the desirable items which, if not available, will not affect the functioning of the hospital.

If one combines the two types of analyses, the results will fall between having a very expensive but desirable drug (which is best avoided) to an ideal, vital product. The balance of the rest of the products falls on a continuum of the above categories.

In retail community pharmacy, stock control is different from hospital pharmacies and will vary from pharmacy to pharmacy. Factors that influence stock holding and control will be related to the following:

- Seasonal factors (e.g., winter vs. summer, rainy season in the tropics, cyclic illnesses/infections)
- Doctors' prescribing habits in the vicinity (e.g., there could be an oncology practice or pain clinic or a dermatologist nearby that could influence the type and quantity of drugs held in stock).
- The size and stock turnover of the pharmacy
- The area that the pharmacy serves (e.g., population ethnicity, affluent/poor) might also affect the price and sale of products held in stock.

Proper storage of goods ensures that the patient receives the highest quality of drug that they deserve. Maintaining the cold chain during transportation, using light-resistant containers, pilfer-proof lids, and securing are all physical measures that pharmacists should implement as part of inventory control measures.

Drugs that should be stored at room temperature means between 15 and 25 or up to 30°C; cool temperature means between 8 and 15°C; refrigeration means between 2 and 8°C; and freezing temperature means −10 to −25°C. Different countries have different room temperatures and definitions of cool temperatures.[15]

Drug distribution in a hospital, from main pharmacies to satellite pharmacies, should be one of the most critical elements of inventory management. Too much stock leads to the risk of wastage and pilferage, and too little may cause patients to experience potentially life-threatening conditions untreated. Different methods of inventory control can be employed, such as the following:[16]

1. Floor stock system—drugs are given to patients through nursing stations; patients are charged for them on administration,
2. Individual prescription or separate billing system—drugs prescribed are billed separately on the patient's unique hospital administration record. Common drugs (e.g., aspirin, acetaminophen) may be included in the hospital room charges.
3. Mobile dispensary or drug cabinet system—a standard list of drugs is kept within dispensing cabinets within the unit and administered to patients on a doctor's order. The charge is recorded by the automated system and is charged to the patient's profile and hospital/ward encounter.
4. A centralized unit-dose drug distribution system—all inpatient drugs are dispensed in unit doses from a central pharmacy. Medications may be dispensed at the time the dose is due to be given to the patient or more likely, in timed batches delivered to the unit prior to the scheduled dose.
5. Decentralized unit-dose dispensing—operated through small satellite pharmacies located on different levels of the hospital. The prescription is sent directly to the pharmacist and then entered in the patient's profile. The pharmacist checks the medication order and technicians fill the prescription; the drugs are then sent to the ward for the nurse to administer the medication.

It should be clear which method(s) is(are) followed, and a standard operating procedure (SOP) should be written in the precise and unambiguous language for everyone to understand, implement, and follow correctly.

Some hospitals have a pneumatic conveyor system for easy distribution of drugs and products to units. The system, if used correctly, can be of great benefit and save many person-hours of staff running between the hospital pharmacy and wards to collect and deliver drugs to the patient. Pharmacy technicians supplying drugs to nursing stations and storing drugs in locked drug cabinets can decrease wastage and theft. In some hospitals, all unit doses are packed in the pharmacy and supplied to the clinics which avoid drug wastage, but this increases labor costs. Software programs available to enhance these processes are helpful, as nurses can see the drug distribution firsthand by tracking each medication order on their computers to have an idea of approximately when they will receive the drugs for their patients.[17]

Drugs that are not available from suppliers because of stock-outs or drugs that are discontinued because of lower demand can then be compounded in the pharmacy. It may also be necessary to compound a drug for a specific patient's need. Fewer products are compounded nowadays, and in some countries, very little compounding is performed as it is difficult to control quality during compounding. The other potential issue is that the compounded product usually has a very short shelf life. Expiration dates on compounded drugs vary because of the content (e.g., 1 day for water-based injectables to 360 days for solid preparations). The volatility of the constituents and the water content plays a big role in the shelf life of the product. The pharmacists can send their product for stability testing to an accredited laboratory if unsure.

Checking expiry dates and proper disposal of drugs are two other important tasks pharmacists are responsible for (e.g., disposal without contaminating groundwater, flushing drugs in the toilet). One of the consequences of flushing hormone tablets down the toilet is that fish in the river could absorb significant amounts of estrogen. Some expired products can be sent back to the supplier, and they, the manufacturers, or the hospital will destroy them in the correct environmentally friendly manner. The concentration of active ingredients in expired drugs will reach a level below which they are deemed ineffective beyond their expiry date, and some might even become toxic (e.g., tetracycline—Fanconi syndrome). The principle of first expired, first out (FEFO) should always be followed when packing stock in shelves (i.e., rotation of stock). The stock should be checked regularly, as well as during dispensing, and if found within 3 months of expiration, it is removed from the shelves and sent back for credit if possible to the supplier or manufacturer.[18]

Organization (Arrangement) of Products in a Hospital Pharmacy

The financial system is of utmost importance in assisting the smooth running of making and receiving payments. It will have to make provisions for issuing invoices for sales, receipts for payments, credit notes, and keeping records for debtors and creditors, all the while being integrated with inventory and dispensing systems.

An inventory/stock control system assists with keeping an optimal level of stock as medication levels are depleted and automatically reorders stock. It should be equipped and programmed to assess, minimum stock levels, lead times (i.e., the time it takes from ordering the stock until it arrives at the pharmacy) and even seasonal trends (e.g., ordering more flu drug in winter) when considering appropriate medication levels for that particular pharmacy.

The correct organization of products in any business is critical to the financial success of the business and is even more important in community pharmacy because drugs have the extra burden of relatively short shelf lives. Correct organization of products is important so that pharmacy staff and consumers can easily and quickly access them.

The front portion of the pharmacy, which is accessible to consumers, will have a different arrangement of products than those in the dispensary. The criteria that apply to the front of the pharmacy shop will have more to do with economics and will take into consideration consumers' shopping patterns (i.e., turning right when entering a store and noticing products that are displayed at eye level first). The market leader in a product category will thus be displayed at eye level with a competitor's products arranged around it; similar products can then be displayed in the same area. A good example is toothpaste, which typically will have mouthwash products, as well as floss and toothbrushes arranged in the same area. Gondola ends (shelf ends) is always regarded as the optimum display, as products are noticed more often by consumers than when these same products are displayed elsewhere. Additionally, companies often pay retailers to have their products displayed in those pharmacy areas fairly large in size.

Having sufficient stock creates the image that the consumer has a wide choice and improves sales. Keeping displays neat and tidy is very important, as well as the fact that the stock rotation needs to happen regularly to keep the consumers' attention—especially the regular visitors—when browsing through the aisles. It is of course just as bad to be overstocked as understocked. A good stock management system, whether it be a computerized model or even a manual system that reflects the number of items sold per period (day/week/month) and the lead times (time from order to delivery), is of critical

financial importance. The arrangement of the products should be what the average consumer will expect to find. For instance, all over-the-counter (OTC) products that have to do with colds/flu or stomach conditions need to be clearly identified by signs to assist consumers/patients with ease of purchases. Systematically identifying and organizing the different categories of medications plays a major role in the financial success of the pharmacy.

The arrangement in the dispensary usually is different as it is not accessible to the consumer/patient and has to do more with the ease of dispensing for the pharmacists and technicians. There are different systems implemented that are sometime alphabetical for all drugs or the therapeutic categories (again arranged alphabetically within the category). Drugs arranged in a specific therapeutic category (e.g., antimicrobials, cardiovascular drugs, antidiabetic drugs, contraceptives, etc.) in one place reduces dispensing errors. All antihypertensives are in one place and a look-alike drug from another therapeutic category is not dispensed by mistake as the pharmacist is concious about the therapeutic category. Drugs are also grouped together by dosage form. Each pharmacy decides what is most convenient for them and in many cases a combinaton of methods are used.

Hospital pharmacist has over the years been the purchasers of drugs and medical devices in the hospital. This role has in many instances been taken over by strategic purchasing units in the hospital that use standardized treatment regimens and use standardized formularies. The pharmacist, however, still has a big role to play in increasing efficiencies in drug utilization and management. In most hospitals, pharmacists are part of the Drug/Pharmacy and Therapeutics Committee for the hospital where they can participate in discussions regarding formulary management systems applicable to that specific hospital, taking into consideration scientific and pharmacoeconomic data.

Hospital drugs are usually arranged for unit stock, outpatient stock, and drugs for the emergency department; often the hospital is also the supplier of stock for nearby primary health clinics. This whole process needs careful planning and follows up to ensure a safe and consistent supply of drugs to patients. The hospital pharmacist is not only expected to be the expert in clinical pharmacy matters but is also responsible for the negotiation, drug management, and budget control, as well as personnel management.[19]

OBJECTIVE 4.3. ILLUSTRATE A MODEL PHARMACY AND DETAIL THE FACILITIES REQUIRED

The main components of a pharmacy care clinic are a pharmacy, a lab, a few consultation rooms for doctors and clinical pharmacists, a nursing unit, and a few ambulatory beds for evaluation or procedures (Fig. 4.4). Patients, typically, would enter through the pharmacy, and if referred to a doctor or clinical pharmacist, enter the clinic area and report to the reception desk. Their information is being recorded, and then they would be assessed by a skilled nurse practitioner and sent to the doctor, clinical pharmacist, or the examination room. The doctor can then take care of the patient either by consultation in his/her room, treat the patient in the procedure room, or assess the patient in the examination room. If blood samples need to be taken, the patient will go to the phlebotomy area; blood or any other samples would be analyzed in the laboratory. Then, the clinical pharmacist assesses the patient's needs and adjusts the treatment, if necessary. The clinic should also have a kitchen for staff and a storage area for use by the pharmacy and the clinic.[20]

A community pharmacy should include a small compounding area; however, it may also be located near/within a hospital pharmacy with a small manufacturing unit (e.g., topical preparations, capsule filling, small tablet punching, parenteral nutrition, IV fluids, ampoule sealing). Some hospital pharmacies have such manufacturing units as an extension to a pharmacy.

Patients have the right to professional treatment by the pharmacist, including the right to privacy. An ideal pharmacy would have a semiprivate consultation area where patients receive information regarding the correct use of their medication and provide the pharmacist personal information necessary for the dispensing of their medication. These semiprivate consultation areas are usually located as part of the dispensing area (Fig. 4.5).[19]

Whenever the situation necessitates, consultations of a more private nature need to take place in a separate private area where confidential information (e.g., HIV status, pregnancy) can be discussed openly with the pharmacist without anyone else (e.g., staff, other clients in the pharmacy) overhearing the discussion. This area usually adjoins the dispensing area. Provisions should also be made for a waiting area for patients, as well as a desk where the pharmacist can be seated for writing purposes, if necessary, and at least two comfortable chairs that can accommodate the elderly and young children (Fig. 4.6).

The third area will be the consultation area which can also serve as the clinic area. This needs to be properly equipped to enable the pharmacist or clinician to perform limited clinical procedures as their respective scope of practice dictates. This area needs an examination couch, a washbasin, and a portable cart with equipment (e.g., stethoscope with baumanometer, otoscope, syringes, needles). There should also be a trauma tray and equipment for the treatment of allergies and other types of minor emergencies. Figs. 4.7 and 4.8 show fully furnished pharmacy areas.

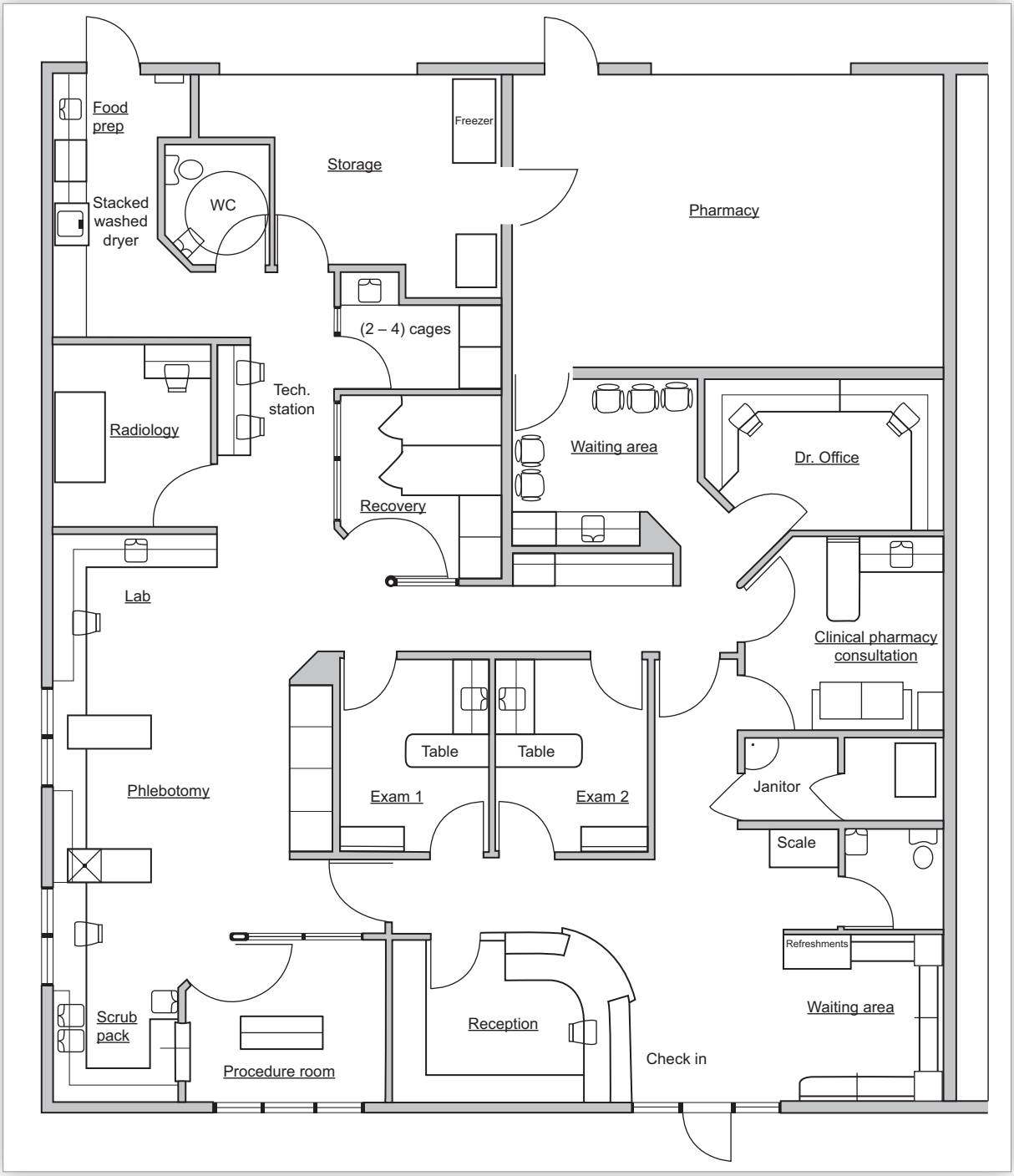


FIGURE 4.4 A model of pharmacy in a clinic.

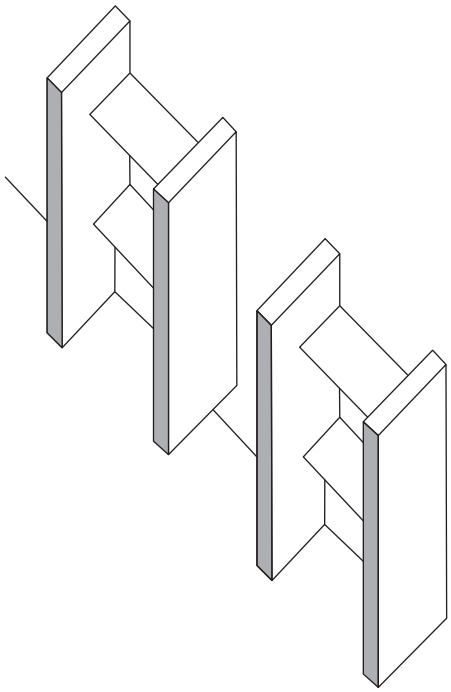


FIGURE 4.5 Semiprivate consultation area.

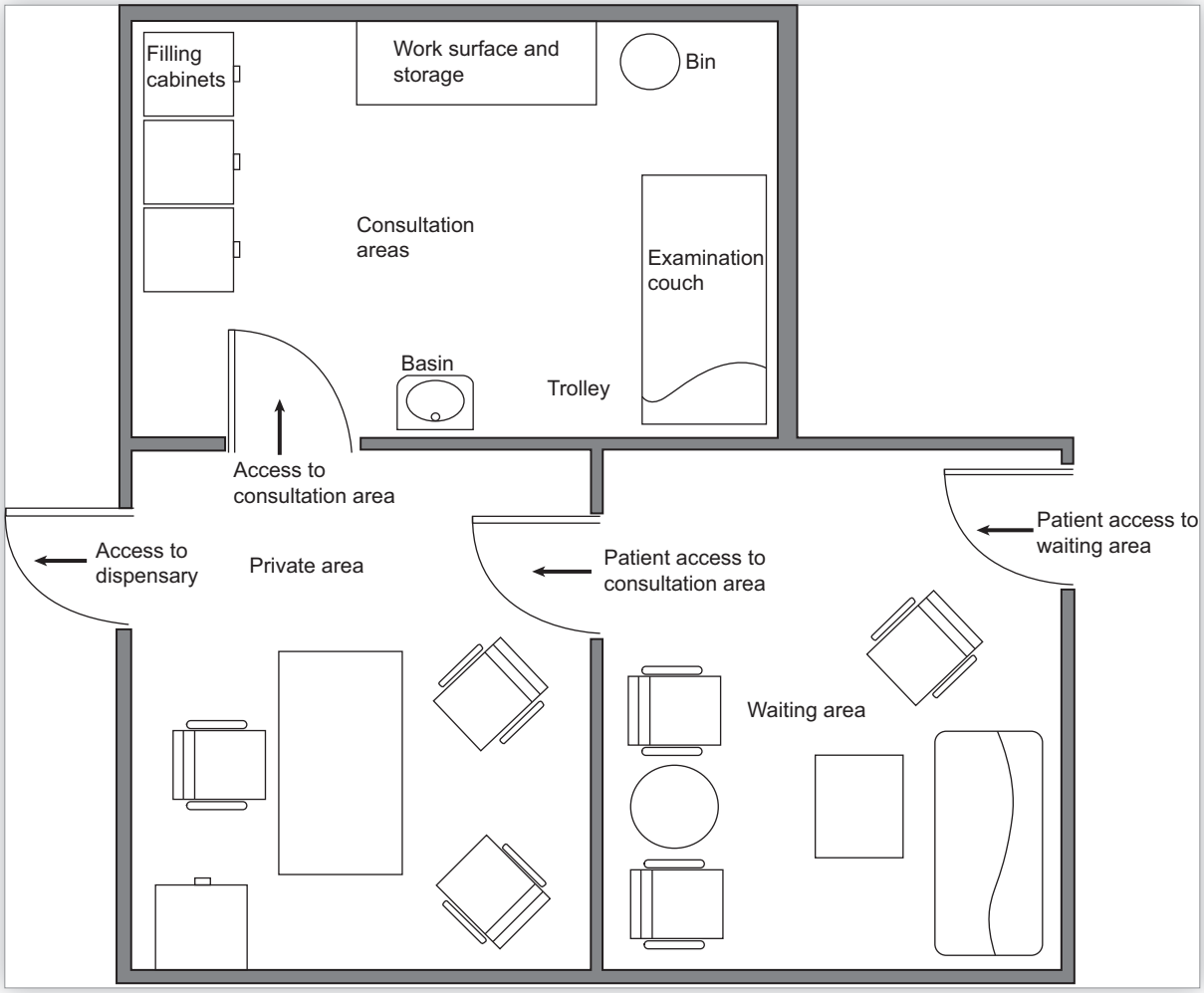


FIGURE 4.6 Private area and consultation room in a pharmacy with large clinical area.



FIGURE 4.7 Inside a pharmacy.



FIGURE 4.8 Pharmacy workstation.

OBJECTIVE 4.4. OVERVIEW PHARMACY PRACTICE SETTING AND RESPONSIBILITIES OF PROFESSIONALS

Pharmacists practice in different work settings, which are popularly known as the following[21–23]:

1. Community pharmacy
   These settings involve improving access to quality drugs and related products to the community it serves. In many countries, still, pharmacists focus heavily on the supply of products and provide little clinical services. This process is

changing globally by providing clinical services appropriate for ambulatory settings. Sometimes called as ambulatory care pharmacies, pharmacists with advanced knowledge and expertise focus on to provide detailed medication consultation. Patients who may have concurrent illnesses and are taking multiple medications either via self-administration or with the help of caregivers need a pharmacist consultation to achieve appropriate clinical outcomes. When not hospitalized and have inconveniences in visiting their treating hospital often, patients prefer to consult pharmacists in community pharmacies. However, the pharmacists should be competent enough to advice in coordination with existing medical care and refer to the hospital if further care is essential. Performing drug interaction screens, as well as medication reconciliations to make sure that the patient's medication list is complete and current is a standard service by the pharmacist. Anticipating adverse outcomes and identifying drug-related problems improves safety of patients. Community pharmacist also treats minor ailments with over-the-counter products, lifestyle modification or with their limited prescribing authority. Thus community pharmacy provides both advanced care in communication with other healthcare faiclities and primary healthcare services independently.

2. Hospital pharmacy (Institutional)
   a. Inpatient pharmacy

      The inpatient pharmacy in a hospital supplies the medicinal needs of patients who have been admitted to the hospital, as well as the drugs that are used during operations and, in most cases, the stock used in the surgery wards, intensive care units, and theatres. It also supplies drug to the general units/wards that are kept as backup stock, or when the pharmacy is closed, it could also be the source of drugs to outlying clinics.

   b. Outpatient pharmacy

      Patients that have been discharged from the hospital, but still need ongoing therapy, are usually supplied by the outpatient pharmacy. They could be patients on chronic medication or supportive medication after treatment (e.g., antibiotics, pain medications, antiinflammatory drugs). Services for outpatients are ambulatory care. Ambulatory care include services for patients not admitted to a healthcare institutions, literally meaning walking patient.
3. Clinical pharmacists

   Clinical pharmacy is a concept that means providing direct patient care services. Irrespective of the site of practice, all pharmacists provide clinical services, such as counseling, prevention of drug-related problems, etc. The pharmacists who work exclusively to provide clinical services are called clinical pharmacists. Clinical pharmacists work in inpatient and ambulatory settings, spending most of their time providing direct patient care but not direct pharmacy dispensing activities. They collaborate with other healthcare professionals to optimize the outcome of drugs in patients. Pharmacy technicians are mostly filling the prescription and the pharmacists are providing exclusive clinical services with dispensing. Pharmacy managers takes care of inventory control. As pharmacy managers and pharmacy technicians performs non-clinical functions of pharmacy, pharmacists could focus fully on clinical services. In this ideal situation, patients are benefited with a full pharmacist interaction that they deserve.
4. Consulting pharmacist

   A consultant pharmacist can operate his/her business by offering a wide range of services visiting the patients. It can range from consulting with senior citizens where the health and medication needs of elderly patients are looked after. In consulting pharmacy, pharmacists visit where the patient lives, especially the old age homes. They differ from other pharmacists who are stationed at a hospital or pharmacy.
5. Managed care pharmacy

   These pharmacies and the pharmacists that work in them are usually attached to health plans and pharmacy benefit management firms. They operate through evidence-based strategies and practices that optimize drug use for their members. They achieve their objective by looking at the most cost-effective treatment for different types of patients primarily by means of drug therapy.

Responsibilities of Personnel Involved in Pharmacy Operations

Apart from the classical categories of pharmacists listed above, there are also the hierarchical categories involved in pharmacy and the ancillary staff involved in a pharmacy.

In a typical hospital pharmacy (and even in community pharmacy) one would also find these different categories of pharmacy staff[19,21,24]:

1. Pharmacy/hospital administrator: To qualify for such a position, pharmacists would usually be required to have at least a Master's degree in Pharmacy or a Master's in Business leadership (or both).

2. Chief pharmacist: This person sees to the proper running of all aspects of the pharmacy (e.g., stock levels, sufficient staff members, organizing staff work schedules); they would have to have a minimum of a Bachelor, Master's degree in Pharmacy, or PharmD. Often, this position is given to a pharmacist who has a significant amount of experience in that particular pharmacy area.
3. Compounding pharmacist: This pharmacist does the specialized compounding required by doctors or hospital staff. They can have specialized training or have received in-house training. A minimum requirement regarding their qualification would need to be a Bachelor in Pharmacy or PharmD.
4. Dispensing pharmacist: This is the most common type of pharmacist found in all areas of pharmacy. They perform all the functions related to the correct dispensing of a prescribed product to a patient including verifying the legality of the prescription, assessing drug−drug, drug−food, or drug−disease interactions, selecting the appropriate drug, and checking for final correctness and expiration dates before handing over the medication to the counseling pharmacist or another clinician. Such pharmacists would need to have a Bachelor in Pharmacy or PharmD as a qualification for this position.
5. Counseling/clinical pharmacist: Their day-to-day tasks can differ from counseling a patient on the correct use of a medication to doing unit/ward rounds with physicians advising on correct regimes, dosages, and contraindications of therapy. These pharmacists are called pharmacotherapy (or clinical) pharmacists, and their training involves a postgraduate course in clinical pharmacy (i.e., PharmD with residency training).
6. Pharmacy intern: These are pharmacy students performing their experiential learning in the hospital/community pharmacy after/during completion of their Bachelor of Pharmacy or PharmD degrees before they can be licensed as pharmacists. They rotate between the different departments of the pharmacy and assist in pharmacy-related tasks allocated to them by their preceptors.
7. Pharmacy technician: This essential personnel performs the technical tasks involved in dispensing medications such as packing medications in their correct places, stocking shelves, and automated cabinets, and general medication preparation tasks freeing the pharmacist to perform other functions of order-entry, verification, and dispensing to ensure patient compliance and medication adherence. In most countries, technicians have to complete a 2-year course at an accredited institution.
8. Administrative/management assistants: These employees see to it that important documentation (e.g., receiving stock, billing to patient records, staff work schedule) and record keeping of SOPs are being done and regularly adhered to. This personnel is usually trained in-house or have a qualification in finances and/or administration.

OBJECTIVE 4.5. EXPLAIN ADVANTAGES AND LIMITATIONS OF PHARMACY AUTOMATION

Automation in pharmacies may decrease the required number of pharmacy technicians and supporting staff, but not necessarily pharmacists. For example, an automatic pill counter improves output and helps to minimize dispensing errors. The same can be said about dispensers, robots, and automated remote dispensers (robots that are controlled from a remote site). Despite their availability, medication error checking and clinical decisions by a pharmacist is considered to be superior to any software. Add to that the counseling of the patient by the pharmacist, as well as other regular tasks performed to decrease medication errors and adverse drug reactions, and automation can be seen as an overall method of improving output by the pharmacist but not replacing them. Compare such advances of automation in other fields that have shown a similar benefit—airplanes have an autopilot, but pilots are still required, and doctors have robots that assist with operations, but it cannot perform the task without being programmed or guided.[21,25]

The advantages of this type of automation are that pharmacy unit-dose dispensing can make a significant impact on the budget/economy within an institution or pharmacy. Automation can assist in compounding/manufacturing, dispensing, and distributing drugs. However, they are costly to acquire, and pricing could be difficult to manage in resource-limited settings.

Pharmacies all over the world make use of computer software to operate dispensing procedures, storage of prescriptions on individual patient profiles, assessment of drug−drug interactions, and evaluate patient-allergic reactions to certain drugs. Software systems can also be used to automatically generate refill prescriptions and communicate with patients regarding the readiness of such refills, as well as provide relevant medical information to the patient (Fig. 4.9).[26]



FIGURE 4.9 Pharmacy automation. *Picture credit: Swisslog Healthcare.*

CONCLUSION

Managing pharmacy needs special skills to manage products, facilities, patients, and staff. Effectively managed pharmacy facilitates access to safe, effective, and affordable drugs and other services by pharmacists. With increased responsibilities to provide patient-based services, it takes more staff time, counseling rooms, and comfort areas for the customers.